INTRODUCTION

This article provides detailed operating instructions for the ITH Stud Tensioning Equipment used to install and remove tensioned casing fasteners. This method applies to MS7001FA gas turbines shipped after September 2000 with tensioned casing fasteners. The tensioning equipment is distributed by ITH Engineering, Inc. for use with tensioned casing studs and nuts supplied by GE.

EQUIPMENT DESIGN INFORMATION

Clamping force on a flange is achieved by tightening the stud or nut until the stud stretches a given length. Standard torqued casing fasteners are normally tightened and loosened by twisting a nut on the stud. This method can result in thread galling, stuck fasteners, and damage to casing flange counterbores. GE Gas Turbines has adopted a hydraulically stretched (tensioned) fastener system for use on selected F-Class compressor, turbine and exhaust casing fasteners. Hydraulically stretched fasteners are designed to minimize galling and disassembly problems by tensioning without twisting the nut or stud. Instead, a hydraulic cylinder assembly engages one end of the stud and hydraulic pressure is applied against the flange to stretch (or tension) the stud. With the stud stretched, the nut is turned down tight against the flange. Once the nut is tight, the pressure is released. The stud remains stretched and held in tension by the nuts, exerting a predetermined clamping force on the flange. For disassembly, the stud is again tensioned, lifting the nut off the flange. The nut is then loosened a few turns, the hydraulic pressure is released, and the nut and stud are easily removed.

REFERENCE DOCUMENTS

	Description
ITH Manual	Bolt Tensioning Equipment Operating Instructions
GE Dwg. 243C1019	Stud, Thread Protectors
GE Standard Practices	

HEALTH AND SAFETY REQUIREMENTS

In all cases, the Health, Safety and Environmental Manual (HSEM) is the document to be followed on the job site during use of this equipment.

Personnel safety and protection of the environment must be an integral part of all business operations, including design, construction, and start-up. Every reasonable effort will be made to maintain a safe and healthful work environment at each customer site.

EQUIPMENT DESCRIPTION

The ITH Stud Tensioning equipment is shipped to the site in a self-contained kit. See Figures 1 through 4. The kit consists of an air-operated hydraulic pump, a set of impact sockets, a set of special Height-Adjustment Sockets (not for impact use), a selection of sized Tensioning Cylinders and Tensioner Supports with attached Gear Drives, ½ inch-drive ratchets, spanner wrenches, and hydraulic hoses.



Figure 1. ITH Stud Tensioning Kit



Figure 2. Tensioning Cylinders, and Tensioner Supports with Gear Drives



Figure 3. Sized ITH Sockets: Height-Adjustment Sockets (left), and Impact Sockets with Flats (right)



Figure 4. Tensioning Cylinder, and Tensioner Support with Gear Drive

PREPARATION FOR ASSEMBLY

Caution: **Several important preparations must be completed before performing the actual stud tensioning procedure with the ITH Tensioning equipment. The following preparatory steps must be performed correctly and in sequence to ensure safe and effective operation of the stud tensioning equipment. Because extremely powerful hydraulic forces will be created during use of this equipment, all operating personnel must fully understand the instructions for operating the tensioner, and become familiar with the cautions and warnings provided to reduce the likelihood of personal injury.**

1. Assembling the Stud and Nut

1. The exposed threads of the stud must be cleaned before installing the tensioning equipment. Clean the threads with a wire wheel or brush and inspect them for damage. Damaged threads must be repaired before tensioning. Apply an anti-seize compound (see Standard Practices for recommended types) to the threads of the stud(s) to be tensioned. Each stud has a “top” (or tensioned) end, where the tensioner is installed, and a “bottom” (or non-tensioned) end, which has an inverted hex used to prevent the stud from rotating during installation. Using the special Height-Adjustment socket, assemble the tensioning nut onto the top of the stud (see Figures 5 and 6). Turn the nut on the stud until the end of the stud lines up with the top of the cutout in the Height-Adjustment socket, and the bottom of the socket lines up with the bottom of the nut (see Figure 7). This establishes the correct amount of thread exposed above the nut needed to attach the Tensioning Cylinder. Assemble the stud and nut in the turbine flange. Assemble another nut onto the bottom of the stud, and run the nut up against the flange hand-tight.



Figure 5. Height-Adjustment Socket: use with Hand Ratchet or Air Ratchet Only – (Not for Impact Use)



Figure 6. Nut Assembled on Stud using Height-Adjustment Socket

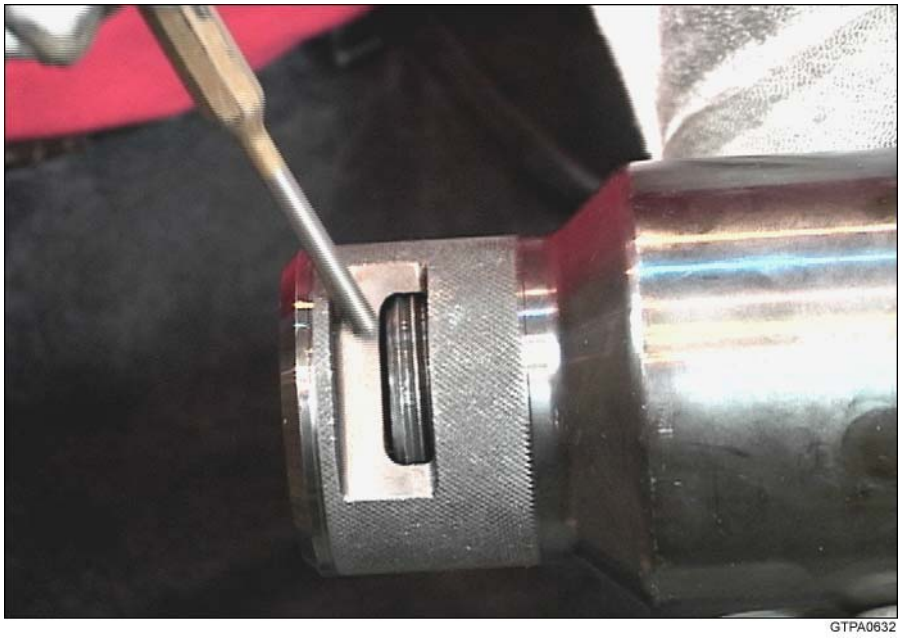


Figure 7. Setting Correct Exposed Thread Length Using Height-Adjustment Socket

2. On the top (tensioned) end of the stud, measure the length of exposed thread above the nut to ensure that there is sufficient thread exposed to operate the Tensioning Cylinder. Establishing the correct amount of exposed thread is important because the Tensioning Cylinder must be fully threaded onto the stud in order to hold it securely while tensioning. To measure, set either the Socket or the Tensioner Support on the exposed surface of the nut on the tensioned end of the stud. See Figures 8, 9 and 10. If the end of the exposed thread falls between the red grooves on the Socket or the Tensioner Support, then the correct amount of thread is exposed for tensioning. Adjust the nut as needed to obtain correct thread length. When correct thread length has been verified, tighten the nut against the flange. Lightly impacting the nut at this point will help prevent the stud from rotating when attaching the Tensioning Cylinder. After tightening, re-verify that the length of exposed thread still falls between the red grooves on the socket or Tensioner Support.



Figure 8. Measuring Exposed Thread Using ITH Socket with Etched Red Grooves

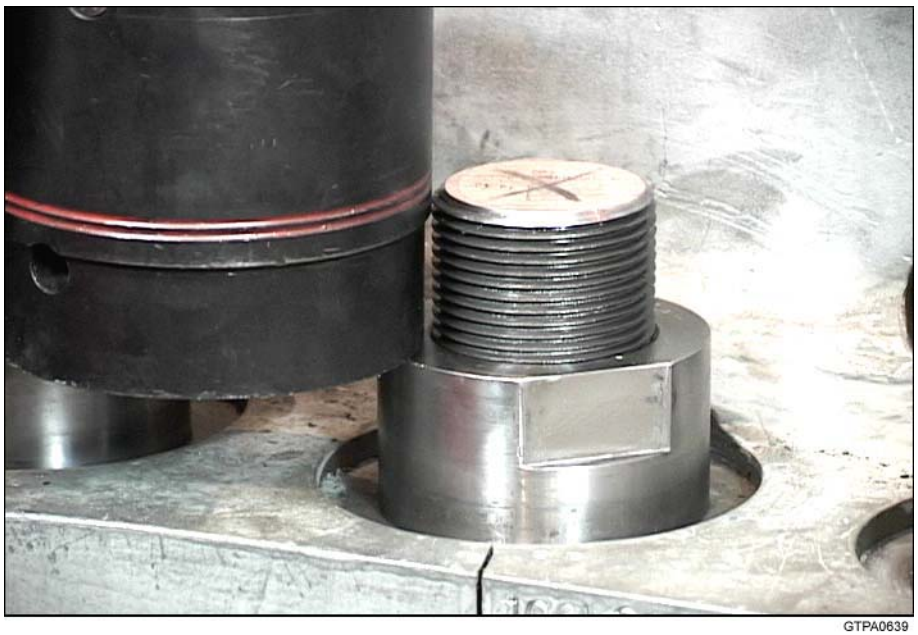


Figure 9. Tensioner Support with Etched Red Grooves

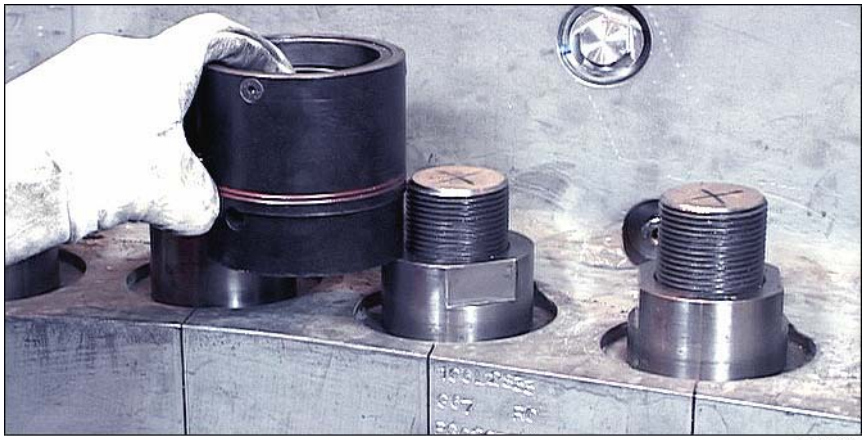


Figure 10. Measuring Exposed Thread Length Using Tensioner Support

2. **Installing the Tensioner**

1. Place the Tensioner Support with attached Gear Drive (Figure 11) onto the nut on the tensioned end of the stud. Ensure that the flats inside the Gear Drive are aligned evenly with the flats of the nut (see Figure 12). If the Gear Drive is properly aligned, the Tensioner Support will be flat against the flange surface.

Note: If the flats of the Gear Drive are not properly aligned with the flats of the nut, the Tensioner Support inner turning sleeve can become damaged during tensioning, resulting in rework and extended assembly time.



Figure 11. Tensioner Support Asssembled on the Nut



Figure 12. Gear Drive Flats Properly Aligned with Flats on Nut

2. Before tensioning, ensure that the Change Bushing on top of the Tensioning Cylinder is down level with the cover, indicating that the piston is in its starting position. Place the Tensioning Cylinder on the Tensioner Support (Figure 13). Using the ½ inch-drive ratchet, thread the cylinder onto the stud by rotating the Change Bushing clockwise (Figure 14). Turn the Change Bushing clockwise until the Tensioning Cylinder is fully threaded onto the stud. Then back off the cylinder about ¼ rotation (90 degrees) to create a small amount of play. This will prevent the tool from binding against the flange during tensioning.

   Note: Ensure that the stud itself does not turn during this step. A hex wrench can be inserted into the ***bottom*** end of the stud to prevent the stud from turning.

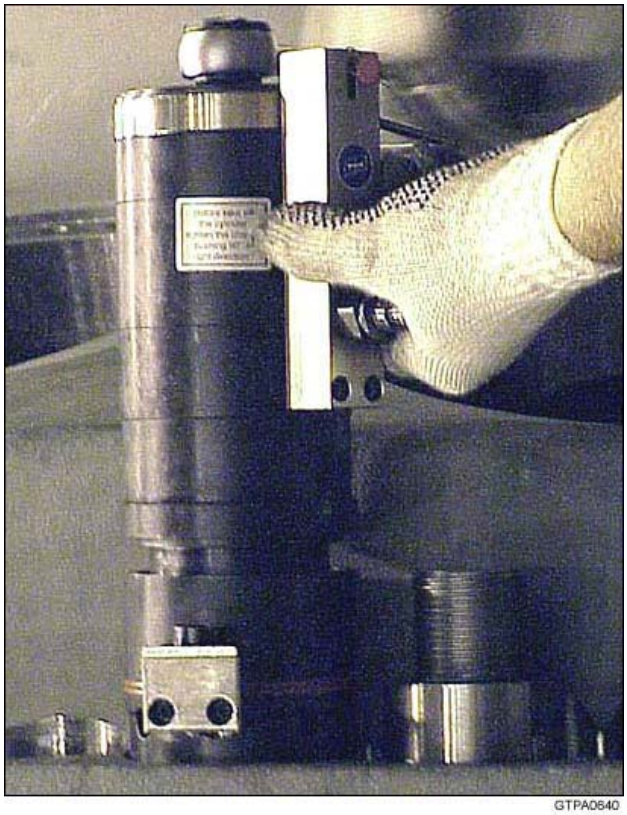


Figure 13. Assembling Tensioning Cylinder onto Tensioner Support



Figure 14. Rotating Change Bushing with Ratchet

3. **Setting the Pump Pressure**

a. A pneumatic pump supplied with the tensioning equipment (Figure 15) is used to create the hydraulic pressure needed for tensioning. This pump requires dry, filtered, compressed air, or an instrument air supply between 60 and 90 psi. Install an air filter-regulator in the supply line if necessary to meet these requirements. Check the internal oil reservoir level before starting the pump. The hydraulic pump pressure **must be set before connecting the pump to the Tensioning Cylinder.** To set pump pressure, first close the shutoff valve. Start the pump, and with the pump running adjust the pressure regulating valve until the needle on the pressure gauge reaches the desired setting. When the desired pressure setting is indicated (e.g., 1350 bar for an MS7001FA turbine), shut off the pump. The next time the pump is started it will generate pressure equal to its last setting. Refer to GE documentation to determine pressure settings for your unit. Do not exceed the recommended pressure for assembly.

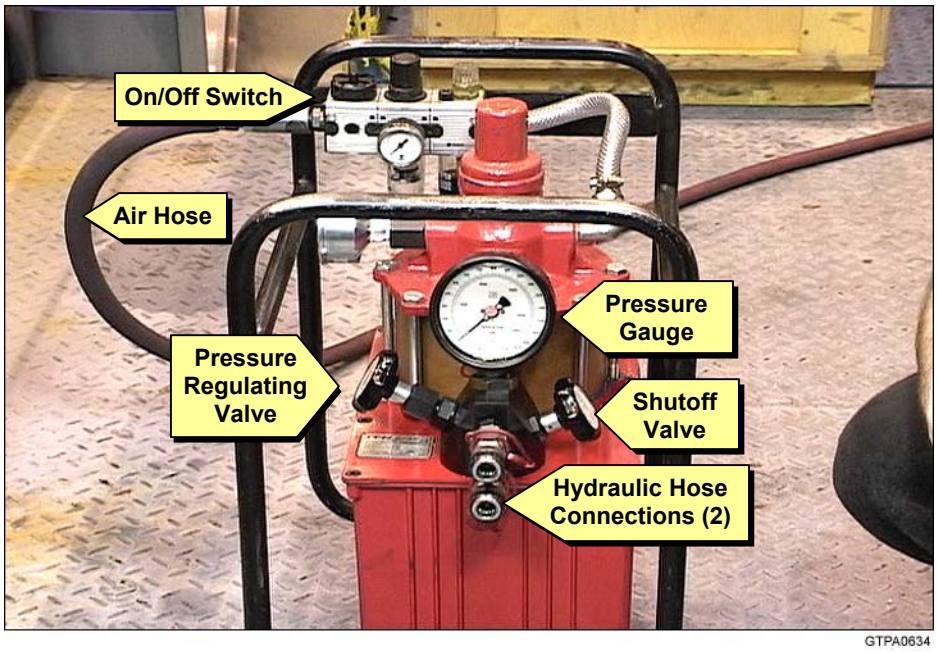


Figure 15. Pneumatic Pump for Creating Hydraulic Pressure

b. Verify that pump pressure is correctly set. Open shutoff valve to release pump pressure and observe that the gauge needle returns to zero. Close shutoff valve and start the pump. The gauge needle should return to the previously selected pressure setting. When correct setting is verified, shut down the pump. Open the shutoff valve to release the hydraulic pressure that has just been created, then close the shutoff valve.

c. Connect one end of the hydraulic hose to the pump and connect the other end to the hydraulic fitting on the Tensioning Cylinder. See Figure 16. Tighten the hose's locking collar to protect against accidental release of the hose connection.

Caution: Before connecting the hydraulic hose to the pump and Tensioning Cylinder, inspect the hose to ensure it is not damaged or kinked. Do not operate with hydraulic hose that is damaged, defective or kinked. A faulty hose may fail under pressure.



Figure 16. Connecting Hydraulic Hose to Tensioning Cylinder

Note: Do not apply pressure to the Tensioning Cylinder unless it is securely threaded onto a stud. Failure to comply can cause the piston to over-stroke, resulting in damage to the Tensioning Cylinder.

With the Tensioning Cylinder fully assembled on the stud, the pump pressure correctly set, and the Tensioning Cylinder connected to the pump, stud tensioning can begin.

STUD TENSIONING PROCEDURE – ASSEMBLY

To ensure personnel safety, the following precautions must be observed by all personnel during this operation.

WARNING: **During stud tensioning, failure of the stud threads or misapplication of the Tensioner may result in the tensioning head becoming a projectile. Personnel must stand clear during tensioning. Except for the operator, all personnel must maintain a distance of at least 10 feet from the equipment while tensioning. The operator must stay clear of being directly above or in line with the tensioning cylinder while it is pressurized.**

1. To begin tensioning, first ensure that the pump shutoff valve is closed. With the tensioning equipment in place on the stud, start the pump and let it run until the reading on the pressure gauge indicates the previously established setting. Then shut down the pump. The Tensioning Cylinder is now stretching the stud so that the nut is no longer in contact with the flange.

2. Using the ½ inch-drive ratchet on the Tensioner Support Gear Drive, rotate the Gear Drive **clockwise** to tighten the nut against the flange surface. See Figure 17. **Remember to use extreme caution when performing this step. Make sure that you are not directly above or in line with the Tensioning Cylinder while it is under pressure.**

   Note: Do not exceed 20 foot-pounds of torque when turning the Gear Drive. Repeated excess torque may result in Gear Drive failure.

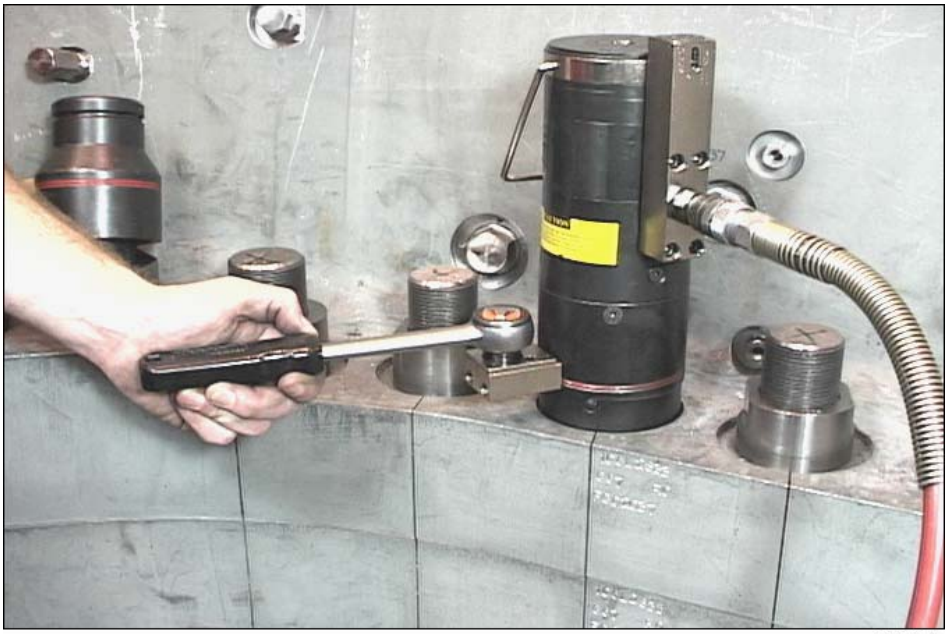


Figure 17. Rotating the Gear Drive

3. When the nut is tight against the flange, open the pump shutoff valve to release the hydraulic pressure in the Tensioning Cylinder. This allows the cylinder's internal springs to return the piston to its starting position.

4. Using the ½ inch-drive ratchet, tighten the Change Bushing on top of the Tensioning Cylinder. This forces any residual hydraulic fluid back into the pump's hydraulic fluid reservoir.

5. Using the ½ inch-drive ratchet, loosen the Change Bushing and remove the Tensioner from the stud. Lift off the Tensioning Cylinder, then lift off the Tensioner Support. The tensioning operation for this stud is now complete.

6. It is now safe to disconnect the hydraulic hose. Be sure to remove the hose from the Tensioning Cylinder **before** disconnecting it from the pump. If planning to tension a number of studs consecutively, the hydraulic hose may remain attached and the entire assembly moved onto the next stud, provided that the hydraulic pressure setting remains the same for the subsequent studs.

7. When tensioning is complete, install thread protectors on the Tensioned End of the stud. Refer to GE dwg. 243C1019, Thread Protector, Stud, for specifications. If thread protectors were not supplied with the unit, contact Product Service for availability.

Caution: **There are 2 hydraulic hose connections on the pump so that two tensioners may be operated simultaneously from a single pump (i.e., one on each side of a casing). Operating two tensioners simultaneously requires careful coordination by the pump operator to ensure that both tensioners are ready before hydraulic pressure is applied.**

STUD TENSIONING PROCEDURE – DISASSEMBLY

Tensioning for disassembly is essentially the reverse of assembly. For disassembly, the stud is again tensioned, lifting the nut off the flange. The nut is then loosened a few turns, the hydraulic pressure is released, and the nut and stud are easily removed. The preparations required for disassembly are also much like the preparations required for assembly.

1. Starting at the tensioned end of the stud, remove the thread protectors. It is extremely important at this point to **clean the exposed threads of the stud before installing the tensioning equipment.** Clean the threads with a wire wheel or brush and inspect them for damage. **Damaged threads must be repaired before tensioning.** Also carefully clean the counterbore around the stud and nut where the tensioner will be seated. Use a wire brush and compressed air to completely clean the surface of paint, dirt and debris.

2. Check the length of exposed thread protruding from the nut by placing either the ITH Socket or the Tensioner Support on the exposed surface of the nut adjacent to the exposed thread of the stud. If the end of the exposed thread falls between the etched red grooves on the Socket or Tensioner Support, then sufficient thread is available for tensioning.

WARNING: **If the length of exposed thread is too short and does not fall between the red grooves, do not attempt to tension the stud. Contact Product Service or the tool manufacturer (ITH) for removal procedures. Attempting to tension without sufficient thread engagement can result in the tensioner head becoming a projectile.**

3. Apply an anti-seize or light oil to the clean, exposed threads. Place the Tensioner Support with Gear Drive onto the nut. As in the assembly procedure, ensure that the flats in the Gear Drive inner turning sleeve are evenly aligned with the flats on the nut (Figure 12). Failure to correctly align the flats will result in damage to the inner turning sleeve. Correct alignment is indicated when the Tensioner Support rests flatly against the flange surface.

4. Assemble the Tensioning Cylinder onto the Tensioner Support. Ensure that the Change Bushing is down level with the cover, indicating that the piston is in its starting position. Using the ½ inch-drive ratchet, thread the cylinder onto the stud by rotating the Change Bushing clockwise. Rotate the Change Bushing until the Tensioning Cylinder is fully threaded onto the stud. Once this is done, loosen the Tensioning Cylinder by backing it off 1 full rotation (360 degrees). This differs from the assembly procedure, in which the cylinder is backed off only 90 degrees. For disassembly, the Tensioning Cylinder **must be backed off 1 full rotation** (360 degrees) to allow enough play for the stud to contract. If there is insufficient play, the tensioning tool will be pulled tightly against the flange which can result in binding of the tool.

5. The hydraulic pressure required for tensioning a stud must be established and set before connecting the pump to the Tensioning Cylinder. To establish pressure setting, first close the pump shutoff valve. Start the pump, and with the pump running adjust the pressure regulating valve until the desired pressure setting is indicated on the pressure gauge. When the desired setting has been attained, shut down the pump. Note that the hydraulic pressure required for tensioning **disassembly** is about **25 to 50 bar greater** than the pressure required for assembly. For example, the pressure required for disassembly of MS7001FA casings is about 1,400 bar. Refer to GE documentation to determine the correct unit-specific hydraulic pressure requirements.

   Caution: Do not exceed 1500 bar of hydraulic pressure.

6. Verify that the pressure setting is correct. Open the shutoff valve to release hydraulic pressure and observe that the gauge needle returns to zero. Close the shutoff valve, restart the pump, and observe that the gauge needle returns to the previously selected setting. When correct setting is verified, shut down the pump. Open the shutoff valve to release the hydraulic pressure that has just been built up, then close the shutoff valve.

7. Connect one end of the hydraulic hose to the pump, then connect the other end to the hydraulic fitting on the Tensioning Cylinder. Tighten the hose's locking collar to protect against accidental release of the hose connection.

   Caution: Before connecting the hydraulic hose, inspect the hose to ensure it is not damaged or kinked. Do not operate with a hydraulic hose that is damaged, defective or kinked. A faulty hose may fail under pressure.

Note: Do not apply pressure to the Tensioning Cylinder unless it is securely threaded onto a stud. Failure to comply can cause the piston to over-stroke, resulting in damage to the Tensioning Cylinder.

8. With the Tensioning Cylinder fully assembled on the stud, the pump pressure correctly set, and the Tensioning Cylinder connected to the pump, the equipment is set to begin tensioning for disassembly.

WARNING: **During stud tensioning, failure of the stud threads or misapplication of the Tensioner may result in the tensioning head becoming a projectile. Personnel must stand clear during tensioning. Except for the operator, all personnel must maintain a distance of at least 10 feet from the equipment while tensioning. The operator must stay clear of being directly above or in line with the tensioning cylinder while it is pressurized.**

9. Ensure that the pump shutoff valve is closed. Start the pump and verify that the gauge indicates the preset pressure. Shut down the pump. The Tensioning Cylinder is now stretching the stud so that the nut is no longer tight against the flange.

10. Using the ½ inch-drive ratchet, rotate the Gear Drive **counterclockwise** 3 complete rotations to loosen the nut (see Figure 17). Three complete rotations of the Gear Drive equal 1 complete rotation of the nut. **Remember to use extreme caution when performing this step. Make sure that you are not directly above or in line with the Tensioning Cylinder while it is under pressure.** Do not exceed 20 foot-pounds of torque when turning the Gear Drive.

11. When the nut is loose, open the pump shutoff valve to release the hydraulic pressure in the Tensioning Cylinder. This allows the cylinder's internal springs to return the piston to its starting position.

12. Using a ratchet in the Gear Drive, rotate the Gear Drive **clockwise** just enough to bring the nut into contact with the flange surface. This will prevent the stud from rotating when removing the Tensioning Cylinder.

13. Tighten the Change Bushing on top of the Tensioning Cylinder to force any residual hydraulic fluid back into the pump's hydraulic fluid reservoir.

14. Using the ratchet, loosen the Tensioning Cylinder from the stud. Remove the Tensioning Cylinder from the Tensioner Support, then remove the Tensioner Support from the nut.

15. Before moving to the next stud, it's a good practice loosen the nut a couple of turns by hand to prevent the nut from retightening as other studs are being tensioned.

16. Disconnect the hydraulic hose. Disconnect the hose from the Tensioning Cylinder **before** disconnecting it from the pump. If tensioning a number of studs consecutively, the hose may remain attached to the Tensioning Cylinder for use on subsequent studs, provided that the hydraulic pressure setting remains the same for those studs.

Troubleshooting Tips For Disassembly

1. During disassembly, if the stud is tensioned but the Gear Drive will not move, try the following actions.

   a) While the stud is tensioned, insert a punch through the access window in the Tensioner Support so that the punch is in contact with the nut. Tap the punch with a hammer to dislodge the nut. See Figure 18.



Figure 18. Inserting Punch in Tensioner Support Access Window to Dislodge Nut

   b) If the Gear Drive still will not rotate, increase hydraulic pressure by 50 bar and attempt loosening again, but in no case should 1500 bar be exceeded.

2. Another troublespot that may occur is difficulty in rotating the Change Bushing while unscrewing the cylinder from the stud. The solution has two simple steps: (1) open the pump shutoff valve, connect a hose to the Tensioning Cylinder and then to the pump. This allows any hydraulic oil trapped in the Tensioning Cylinder to escape back to the pump, (2) tighten down the Tensioning Cylinder and turn down the Change Bushing (clockwise) to squeeze out any remaining oil trapped in the cylinder. When oil pressure on the Change Bushing has been relieved, the Change Bushing should turn normally again.

MS7001FA 4-WAY JOINT – ASSEMBLY NOTES

It is important to recognize the following concern pertaining to assembly of the MS7001FA casing 4-way joint between the Compressor Discharge Case (CDC) and the turbine case. After the 4-way joint has been sufficiently pulled together, and the other horizontal and vertical bolting has been tensioned in the usual sequence, final tensioning of the 4-way joint must be performed in the order described here. Failure to follow this sequence can result in interference between the tensioner and other studs.

1. Figure 19 illustrates the 4-way joint tensioning sequence. The 4 outer horizontal studs and the 4 outer vertical studs at the joint should be tensioned first **[1]**.

2. Tension the first vertical joint stud above the 4-way flange **[2]**. Select the “Deep-Well” Tensioner Support designed specifically for this location, and tension from the ***turbine case*** (aft) side of the vertical flange.

3. Next, the adjacent studs on the horizontal joint can be assembled and tensioned: 3 studs aft **[3]** and 1 stud forward **[4]**. The tensioner should go ***above,*** not below, the joint.

4. Finally, the next 2 vertical studs above the joint **[5]** should be tensioned from the ***forward*** side of the flange.

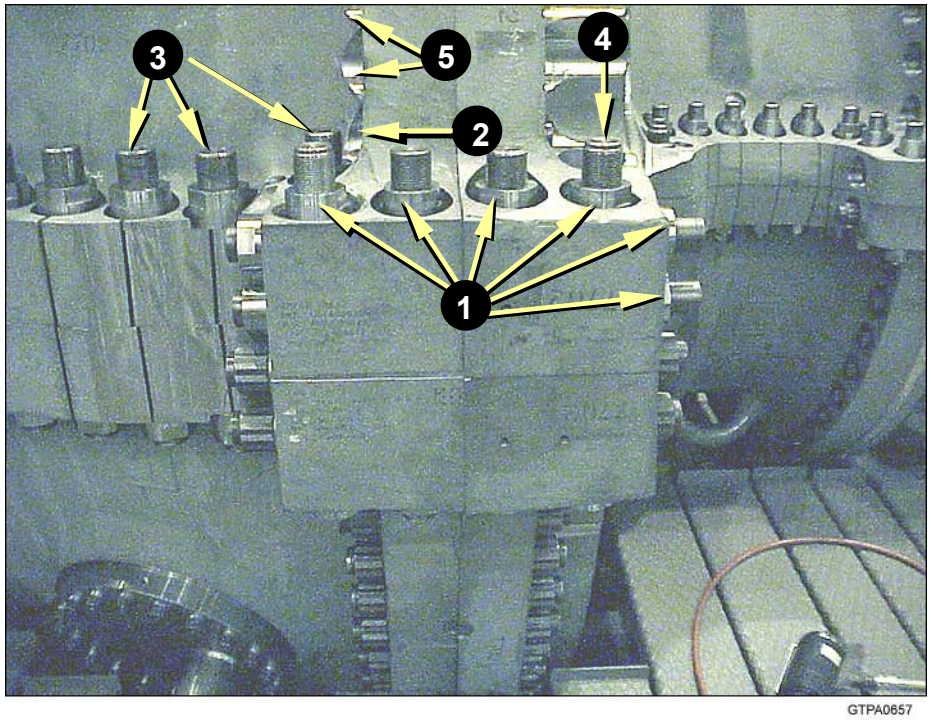


Figure 19. 4-Way Joint Tensioning Sequence (MS7001FA)

SPECIAL ASSEMBLY INSTRUCTIONS – MS7001FA CDC INNER GAS PATH STUDS

It is also important to note that tensioned hardware in the CDC inner gas path must be locked with a secondary retainer nut and staked on both ends. The tensioned studs in these locations (Figure 20) are assembled with 3 nuts: 1 regular tensioning nut and 1 slotted retainer nut on the top, and 1 slotted tensioning nut on the bottom (Figure 21). The staking procedure is not provided on the drawing and **must be** done in the order described here.

1. Run the slotted lower nut onto the bottom (hex) end of the stud until one thread extends past the bottom of the nut. Stake each stud in a slot at two locations. Do this on a workbench or table for each of the 14 studs before they are installed in the casing. Refer to Figure 22. Do not install the studs in the inner gas path until the lower nuts have been staked. Verify that the nuts cannot back off.



Figure 20. Tensioned Inner CDC Stud Locations

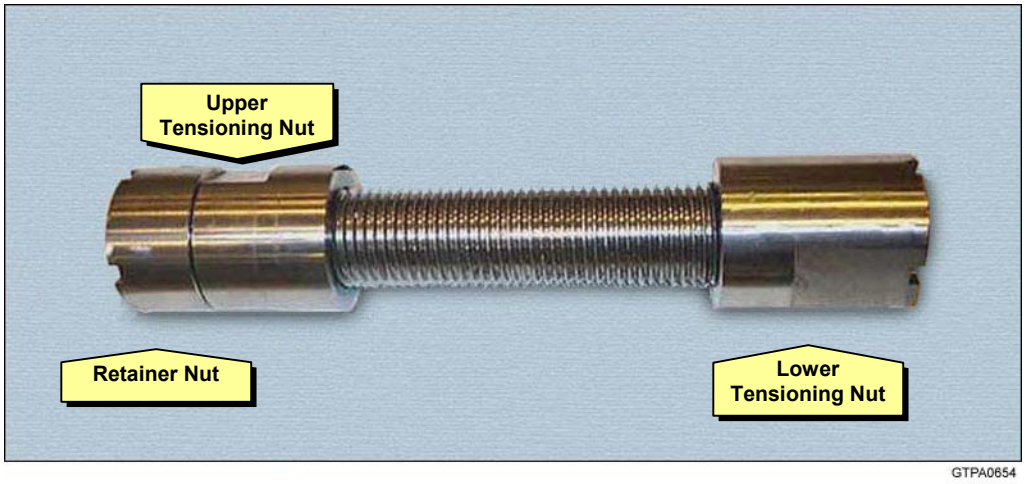


Figure 21. Stud and Nuts Used for Inner CDC



Figure 22. Staking Studs Before Assembly into CDC

2. Install the 14 studs (with the lower nut already staked) into the 7 forward bolt holes on the inner CDC flange right and left sides. See Figure 23.

3. Install the top tension nut on each stud. Impact lightly to tighten down the nut, then verify correct thread height above each top tension nut.

4. Proceed to tension all the studs (typically 14 tensioned studs – 7 on each side of the inner gas path on an MS7001FA).

5. After tensioning is complete, run the retainer nut down on each of the studs (see Figure 24). If the nut has no flats, it can be tapped tight using a pin or punch in a slot. On later units, the retainer nuts have 2 flats and can be tightened with a wrench.

6. Stake the retainer nut to the stud threads in two slot locations. Side impact stake. See Figure 25.

7. Verify that the nut cannot back off.

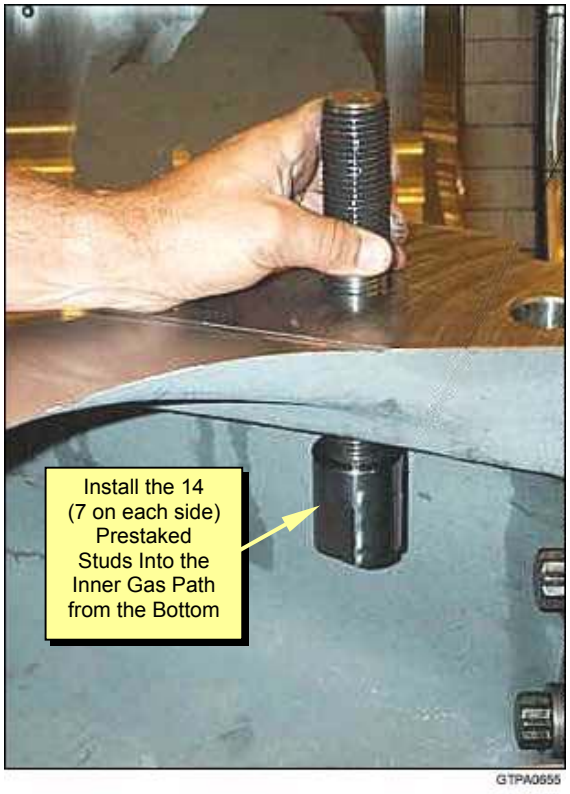


Figure 23. Installing Stud in Forward Bolt Holes

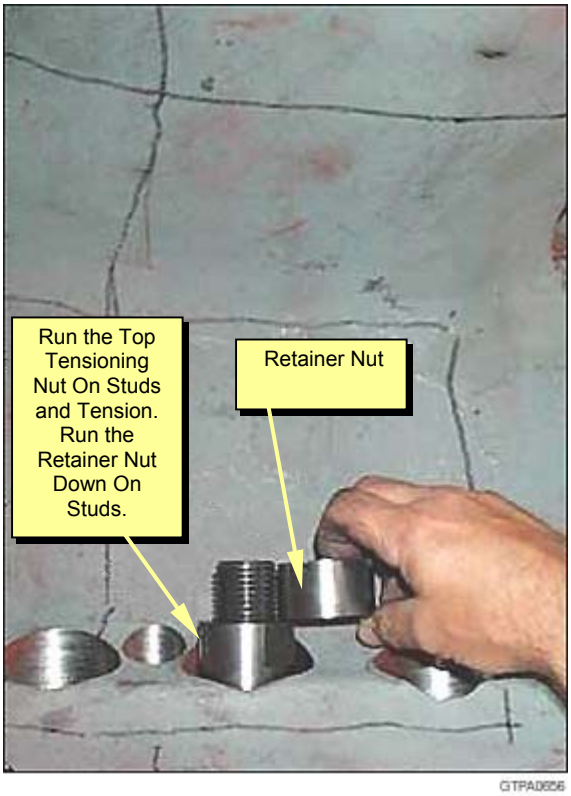


Figure 24. Installing Retainer Nut



Figure 25. Staking the Top Retainer Nut After Installation

SPECIAL DISASSEMBLY INSTRUCTIONS – MS7001FA CDC INNER GAS PATH STUDS

When removing the CDC inner gas path studs, first remove the top retainer nut (which was staked during assembly). Locate the stakes if possible and, using a small burr, clean up the metal on the staked area of the thread. A small die grinder with a cone-shaped carbide burr may work. Early versions of the retainer nuts have no flats: use a pipe wrench to remove them. Later versions of the retainer nut have two flats, so they can be removed with an impact or socket wrench. Once the retainer nut is removed, the tensioned nut can be accessed. **Before installing the Tensioning Cylinder, a thread chaser should be used to clean up the threads where the staking occurred**. Install the tensioner, and proceed to tension and disassemble the upper nut per procedure. The stud and lower nut can then be removed as an assembly. There is no need to remove the lower nut (which is also staked) from the stud.

WARNING: **During stud tensioning, failure of the stud threads or misapplication of the Tensioner may result in the tensioning head becoming a projectile. Personnel must stand clear during tensioning. Except for the operator, all personnel must maintain a distance of at least 10 feet from the equipment while tensioning. The operator must stay clear of being directly above or in line with the tensioning cylinder while it is pressurized.**

EQUIPMENT CARE AND STORAGE

1. Check the pump's internal oil reservoir level before beginning work.

2. This pump requires dry, filtered, compressed air, or an instrument air supply between 60 and 90 psi for operation. Install an air filter-regulator in the supply line if needed to meet these requirements.

3. The Stud Tensioning Cylinder is equipped with a cycle counter, located on the cylinder's hydraulic fitting block, to record the number of cycles the tensioner has performed. Stamped on the fitting block is the maximum number of cycles the tensioner can perform for safe operation before the Change Bushing must be replaced. After completion of work, note and record the number on the cycle counter to determine if the tensioning cylinder is nearing the end of a scheduled maintenance interval. See Figure 26.

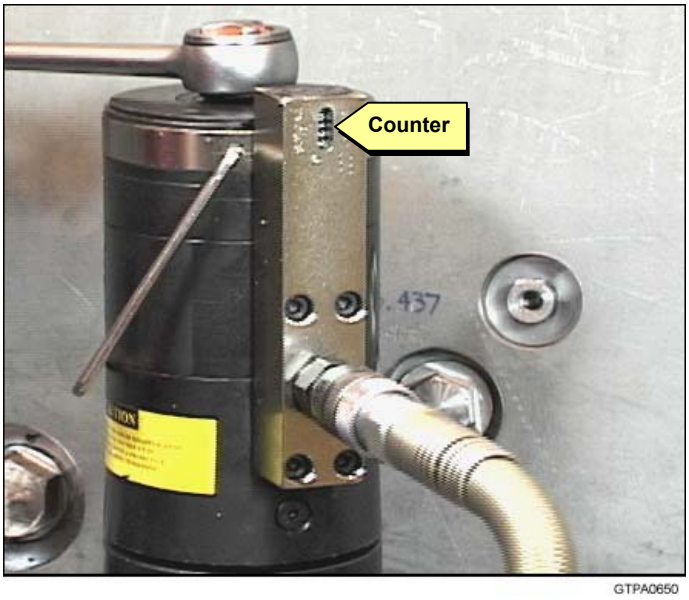


Figure 26. Cycle Counter on Tensioning Cylinder Hydraulic Fitting Block

4. When storing the pump after completion of work, be sure to loosen the pressure regulating valve on the pump. This will help increase the life of the valve's internal spring.

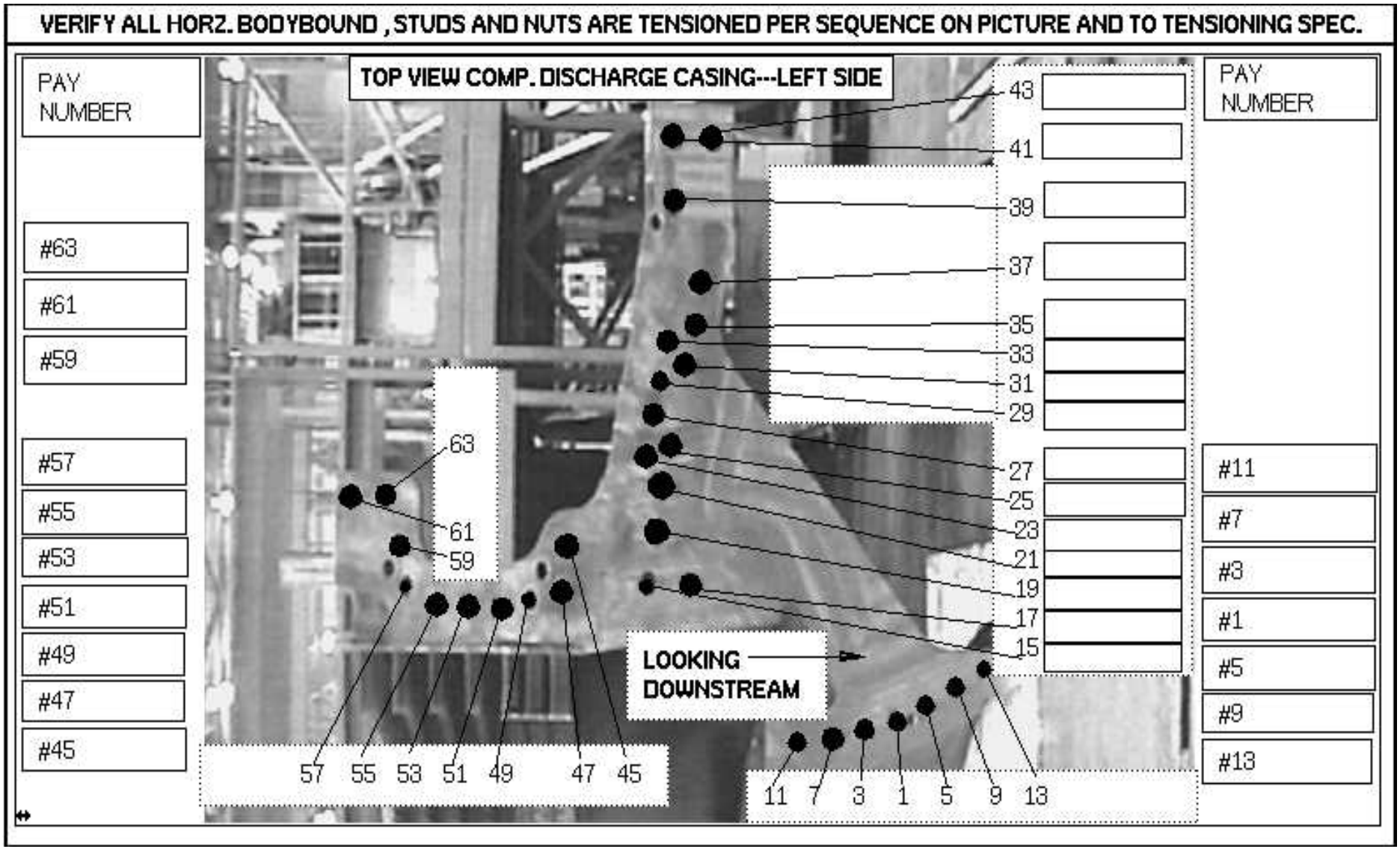
5. Dry, climatized storage is essential for proper functioning of the unit.

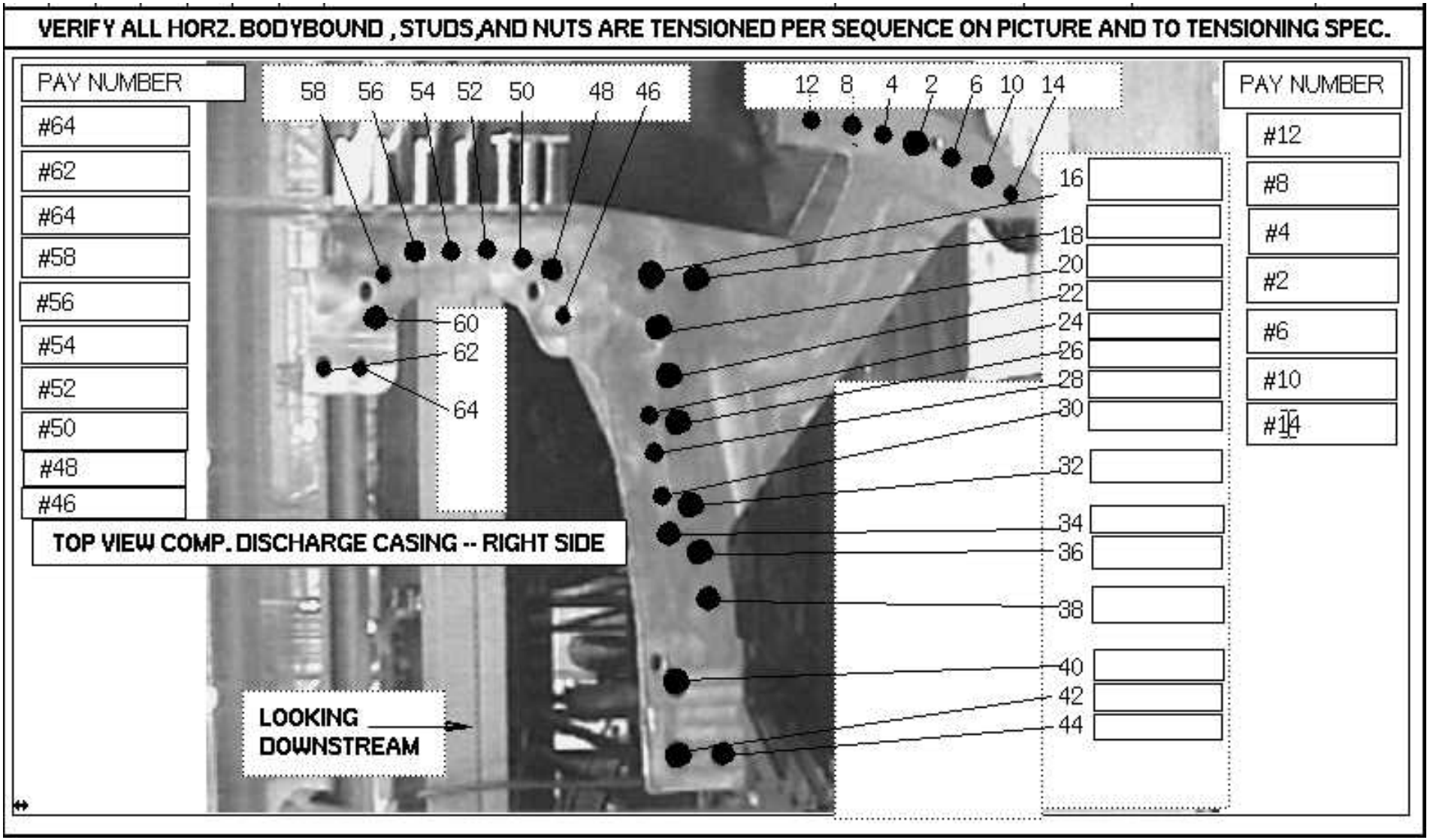
IMPORTANT POINTS TO REMEMBER

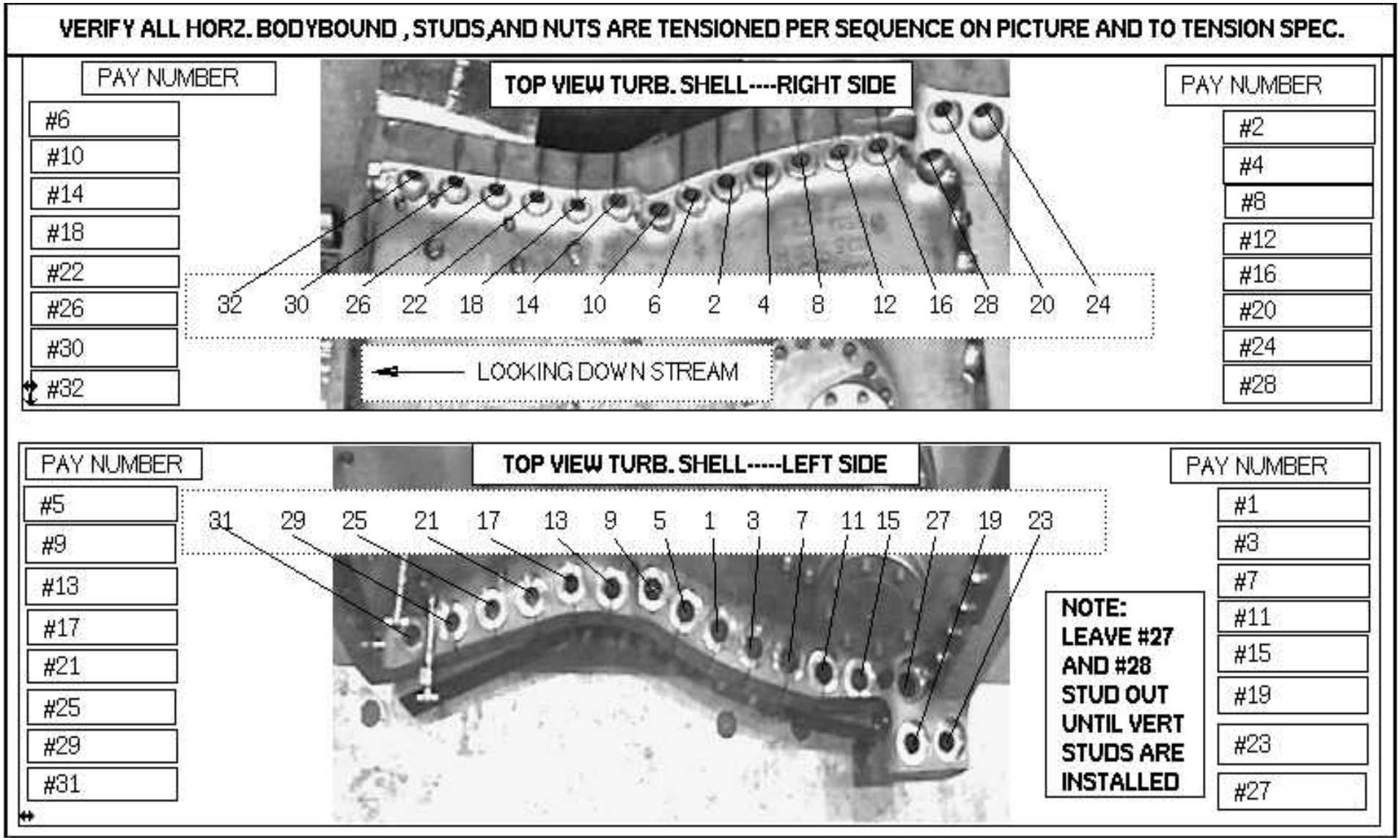
1. Review the manufacturers instructions carefully before starting work. Refer to GE documentation for unit-specific settings.
2. Always wear safety goggles when working with hydraulic equipment.
3. Inspect hoses and fittings before use. Do not use a hose that is damaged or kinked.
4. Clean and inspect stud threads before threading tensioner onto stud.
5. Clean the counterbore around the nut before tensioning. The Tensioner Support will not rest flatly against the casing surface if there is any dirt or debris.
6. Be sure that the flats inside the Tensioner Support line up with the flats on the nut.
7. Before screwing the Tensioning Cylinder onto the stud, ensure there is sufficient thread above the nut: check thread height against the red grooves. The tensioning tool must be able to grab enough exposed thread to hold the stud securely during tensioning.
8. Check the cycle counter. The maximum number is 2000 cycles.
9. Never apply pressure to a Tensioning Cylinder that is not fully threaded onto a stud.
10. Before tensioning, ensure that the Change Bushing is down level with the cover. Do not start tensioning with the Change Bushing sticking up above its starting position.
11. Never exceed the maximum stroke, indicated by the appearance of a red groove on the Change Bushing. If the red groove appears, immediately stop tensioning and release pressure by switching off the pump and opening the shutoff valve.
12. Except for the operator, all personnel must stand clear and keep a distance of at least 10 feet from the equipment while tensioning is taking place.
13. The operator must not be directly above or directly in line with the Tensioning Cylinder when it is under pressure.
14. Operating two tensioners simultaneously from a single pump requires careful coordination by the pump operator to ensure that <u>both</u> tensioners are ready before hydraulic pressure is applied.
15. Never exceed the Tensioning Cylinder's maximum operating pressure. Maximum operating pressure is given in the manufacturer's instructions.

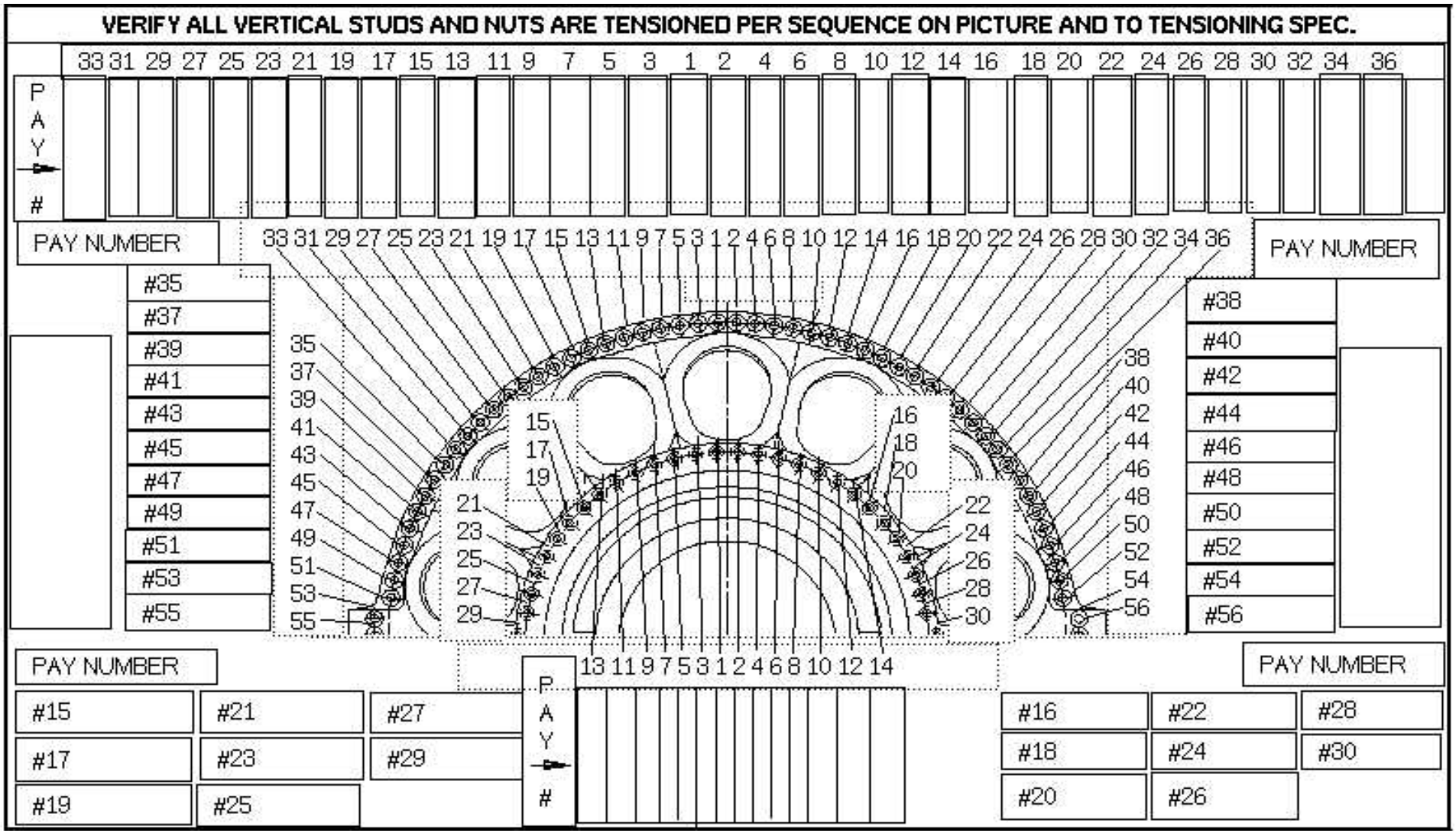
VERIFICATION OF TENSIONING SEQUENCE

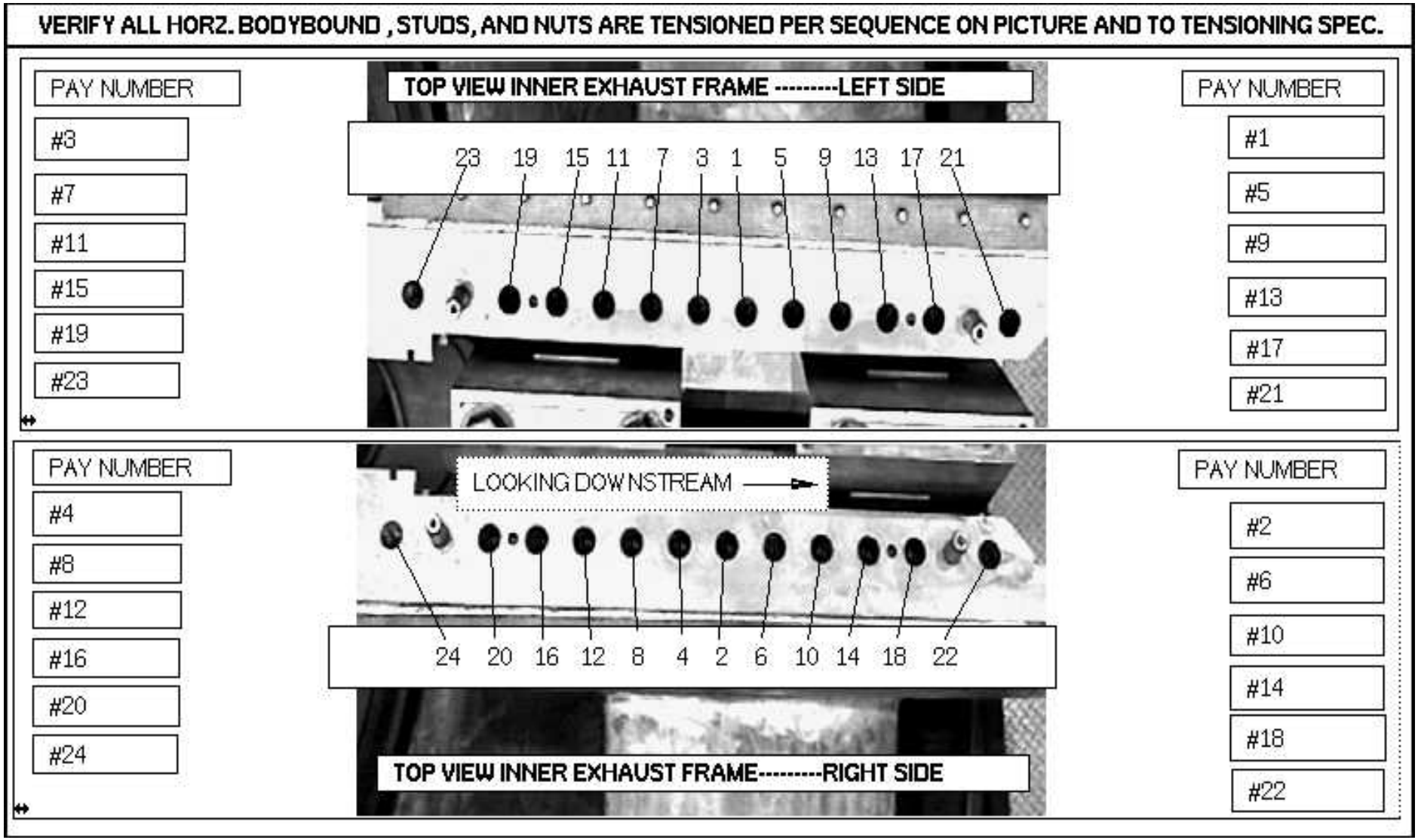
The following illustrations provide the proper sequence for CDC, Turbine Shell and Exhaust Frame tensioning. Operators are required to initial the numbered boxes provided on the illustrations to verify that the studs have been tensioned in the stated sequence.

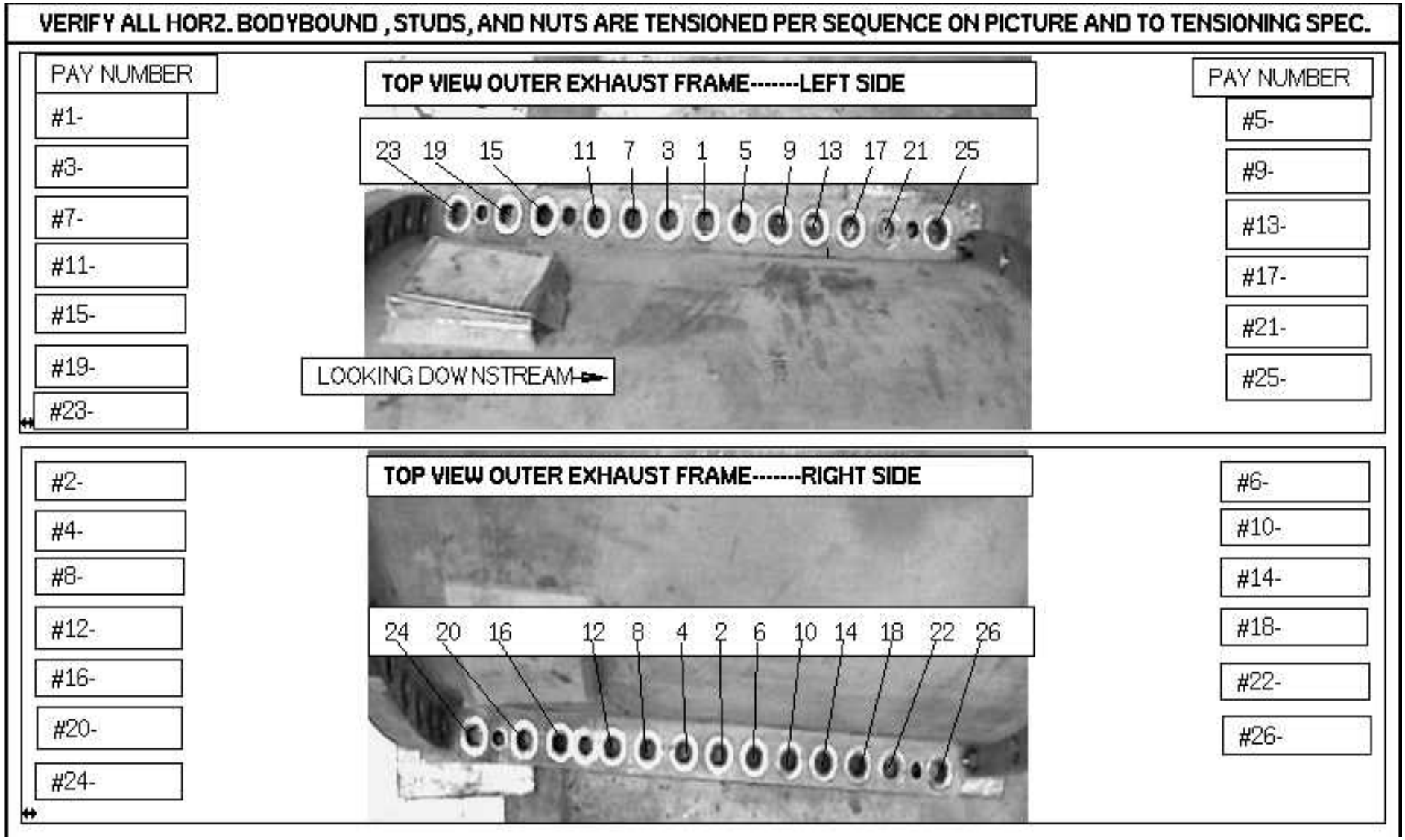

VERIFY ALL HORZ. BODYBOUND , STUDS AND NUTS ARE TENSIONED PER SEQUENCE ON PICTURE AND TO TENSIONING SPEC.
TOP VIEW COMP. DISCHARGE CASING---LEFT SIDE
PAY NUMBER
#63
#61
#59
#57
#55
#53
#51
#49
#47
#45
PAY NUMBER
#11
#7
#3
#1
#5
#9
#13
LOOKING DOWNSTREAM
43
41
39
37
35
33
31
29
27
25
23
21
19
17
15
63
61
59
57 55 53 51 49 47 45
11 7 3 1 5 9 13

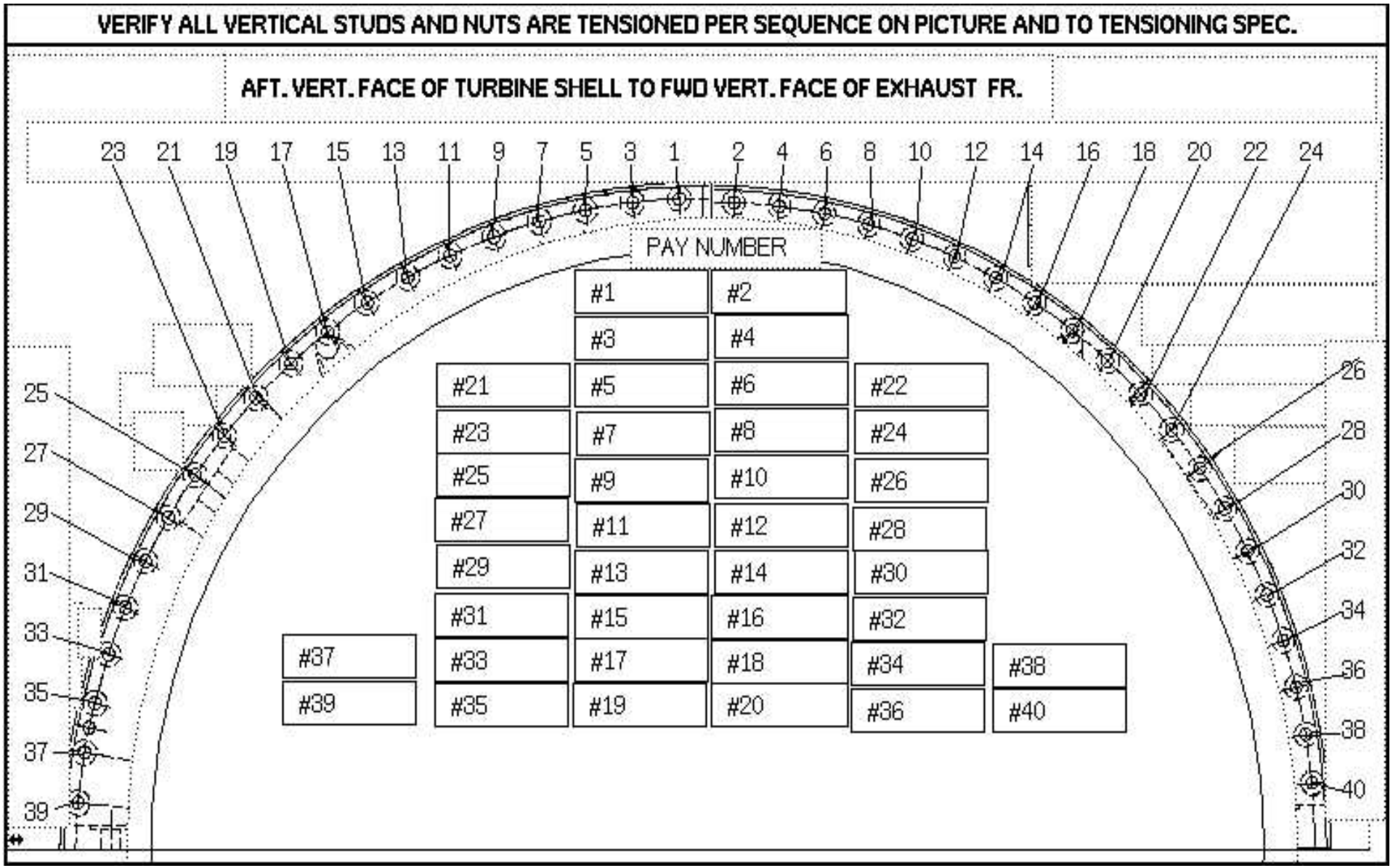

VERIFY ALL HORZ. BODYBOUND , STUDS,AND NUTS ARE TENSIONED PER SEQUENCE ON PICTURE AND TO TENSIONING SPEC.
PAY NUMBER
#64
#62
#64
#58
#56
#54
#52
#50
#48
#46
58 56 54 52 50 48 46
60
62
64
TOP VIEW COMP. DISCHARGE CASING -- RIGHT SIDE
LOOKING DOWNSTREAM
12 8 4 2 6 10 14
16
18
20
22
24
26
28
30
32
34
36
38
40
42
44
PAY NUMBER
#12
#8
#4
#2
#6
#10
#14


VERIFY ALL HORZ. BODYBOUND , STUDS,AND NUTS ARE TENSIONED PER SEQUENCE ON PICTURE AND TO TENSION SPEC.
PAY NUMBER
#6
#10
#14
#18
#22
#26
#30
#32
TOP VIEW TURB. SHELL----RIGHT SIDE
32 30 26 22 18 14 10 6 2 4 8 12 16 28 20 24
LOOKING DOWN STREAM
PAY NUMBER
#2
#4
#8
#12
#16
#20
#24
#28
PAY NUMBER
#5
#9
#13
#17
#21
#25
#29
#31
TOP VIEW TURB. SHELL-----LEFT SIDE
31 29 25 21 17 13 9 5 1 3 7 11 15 27 19 23
NOTE:
LEAVE #27
AND #28
STUD OUT
UNTIL VERT
STUDS ARE
INSTALLED
PAY NUMBER
#1
#3
#7
#11
#15
#19
#23
#27


VERIFY ALL VERTICAL STUDS AND NUTS ARE TENSIONED PER SEQUENCE ON PICTURE AND TO TENSIONING SPEC.
33 31 29 27 25 23 21 19 17 15 13 11 9 7 5 3 1 2 4 6 8 10 12 14 16 18 20 22 24 26 28 30 32 34 36
PAY #
PAY NUMBER
#35
#37
#39
#41
#43
#45
#47
#49
#51
#53
#55
35 37 39 41 43 45 47 49 51 53 55
15 17 19 21 23 25 27 29
16 18 20 22 24 26 28 30
38 40 42 44 46 48 50 52 54 56
PAY NUMBER
#38
#40
#42
#44
#46
#48
#50
#52
#54
#56
PAY NUMBER
#15
#17
#19
#21
#23
#25
#27
#29
13 11 9 7 5 3 1 2 4 6 8 10 12 14
PAY #
PAY NUMBER
#16
#18
#20
#22
#24
#26
#28
#30


VERIFY ALL HORZ. BODYBOUND , STUDS, AND NUTS ARE TENSIONED PER SEQUENCE ON PICTURE AND TO TENSIONING SPEC.
PAY NUMBER
#3
#7
#11
#15
#19
#23
TOP VIEW INNER EXHAUST FRAME ---------LEFT SIDE
23 19 15 11 7 3 1 5 9 13 17 21
PAY NUMBER
#1
#5
#9
#13
#17
#21
PAY NUMBER
#4
#8
#12
#16
#20
#24
LOOKING DOWNSTREAM
24 20 16 12 8 4 2 6 10 14 18 22
TOP VIEW INNER EXHAUST FRAME---------RIGHT SIDE
PAY NUMBER
#2
#6
#10
#14
#18
#22


VERIFY ALL HORZ. BODYBOUND , STUDS, AND NUTS ARE TENSIONED PER SEQUENCE ON PICTURE AND TO TENSIONING SPEC.
PAY NUMBER
#1-
#3-
#7-
#11-
#15-
#19-
#23-
TOP VIEW OUTER EXHAUST FRAME-------LEFT SIDE
23 19 15 11 7 3 1 5 9 13 17 21 25
LOOKING DOWNSTREAM
PAY NUMBER
#5-
#9-
#13-
#17-
#21-
#25-
#2-
#4-
#8-
#12-
#16-
#20-
#24-
TOP VIEW OUTER EXHAUST FRAME-------RIGHT SIDE
24 20 16 12 8 4 2 6 10 14 18 22 26
#6-
#10-
#14-
#18-
#22-
#26-


VERIFY ALL VERTICAL STUDS AND NUTS ARE TENSIONED PER SEQUENCE ON PICTURE AND TO TENSIONING SPEC.
AFT. VERT. FACE OF TURBINE SHELL TO FWD VERT. FACE OF EXHAUST FR.
PAY NUMBER
#1
#2
#3
#4
#21
#5
#6
#22
#23
#7
#8
#24
#25
#9
#10
#26
#27
#11
#12
#28
#29
#13
#14
#30
#31
#15
#16
#32
#37
#33
#17
#18
#34
#38
#39
#35
#19
#20
#36
#40
23 21 19 17 15 13 11 9 7 5 3 1 2 4 6 8 10 12 14 16 18 20 22 24
25 27 29 31 33 35 37 39
26 28 30 32 34 36 38 40